DESIGN PROJECT

DESIGN & ANALYSIS OF DISC BRAKE ROTOR

CONTENTS

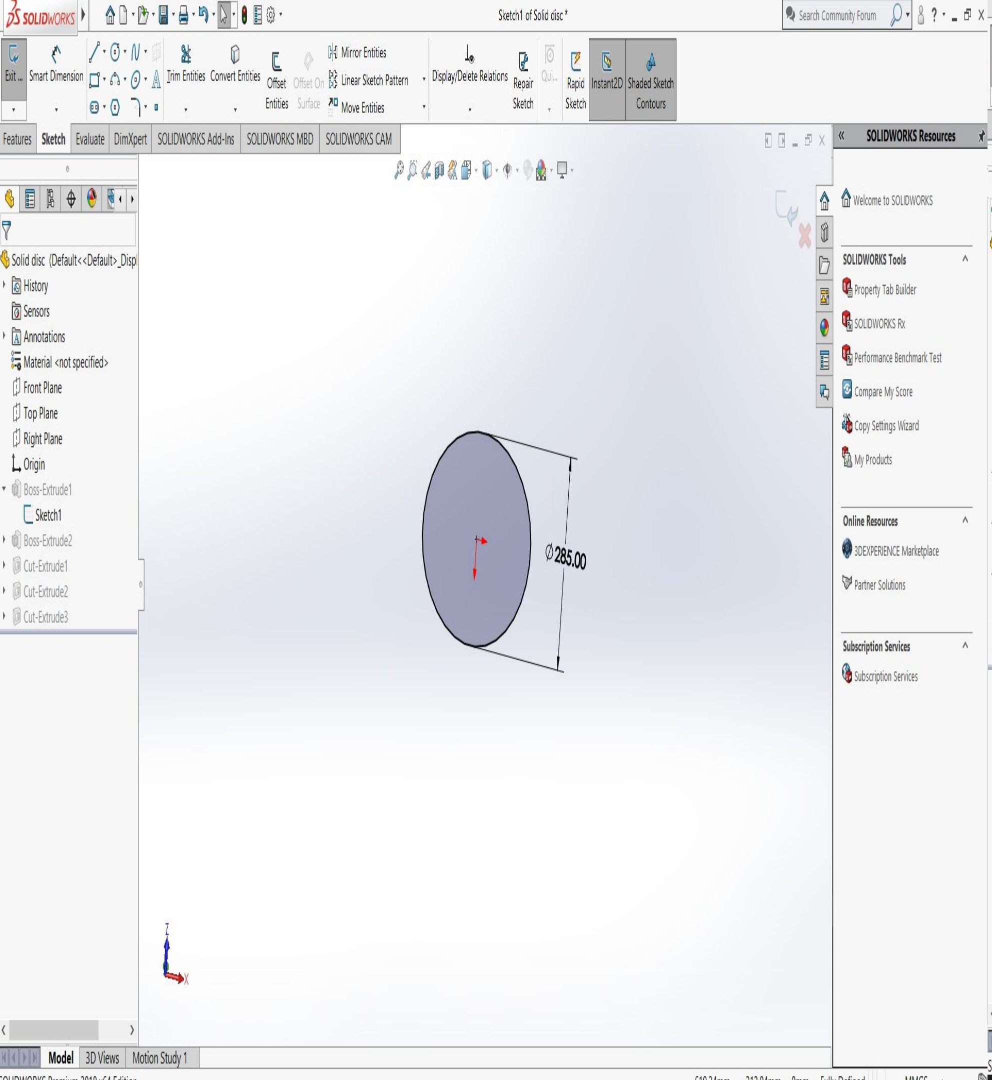
OBJECTIVE

DESIGN AND ANALYSIS OF DIFFERENT TYPES OF DISC ROTORS

INTRODUCTION

Brakes are essential part of a automobile which aids us in contolling the vehicle for a safe riding. A brake is a mechanical device which restrains the motion by absorbing energy from a moving system.The hindrance is often achieved by means of friction.For over a century ,braking systems have evolved into a more complex device to adapt to different road conditions. As technology of automobiles has advanced various forms of brakes have been developed, a Disc brake one such kind. The disc rotor is the rotating part of disc brake assembly,against which the brake pads are applied. In this study design and analysis of different types of disc rotors,which are commonly used in the automobile industry was conducted and the results were compared.

CAD MODELLING



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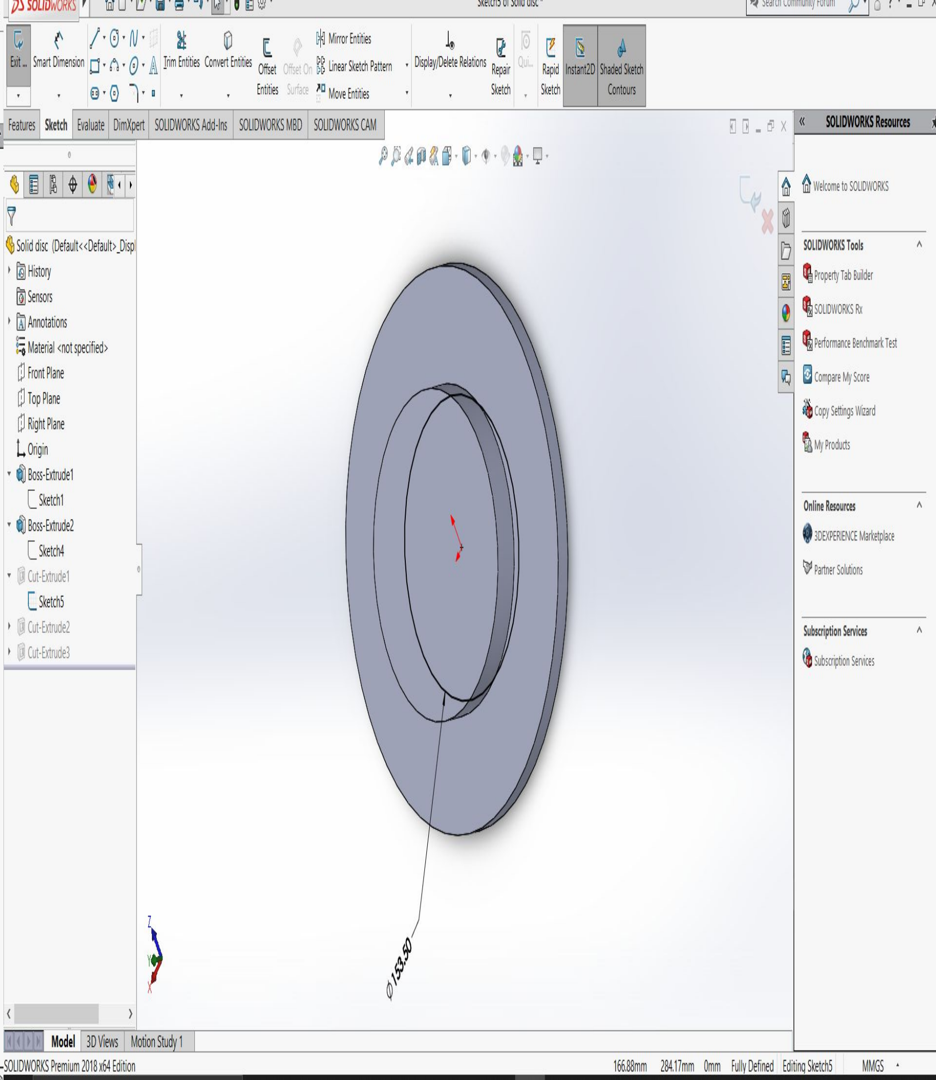

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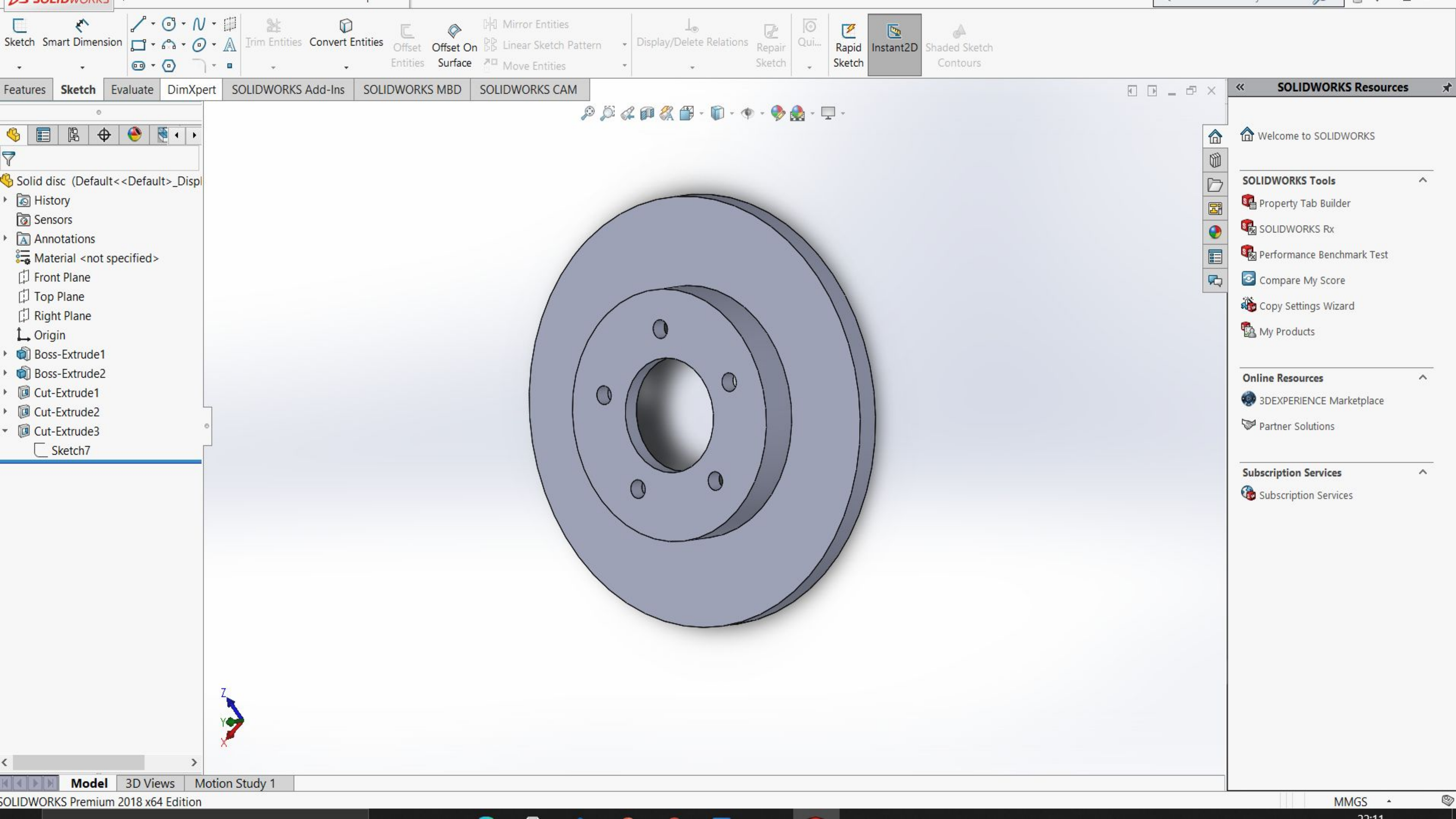
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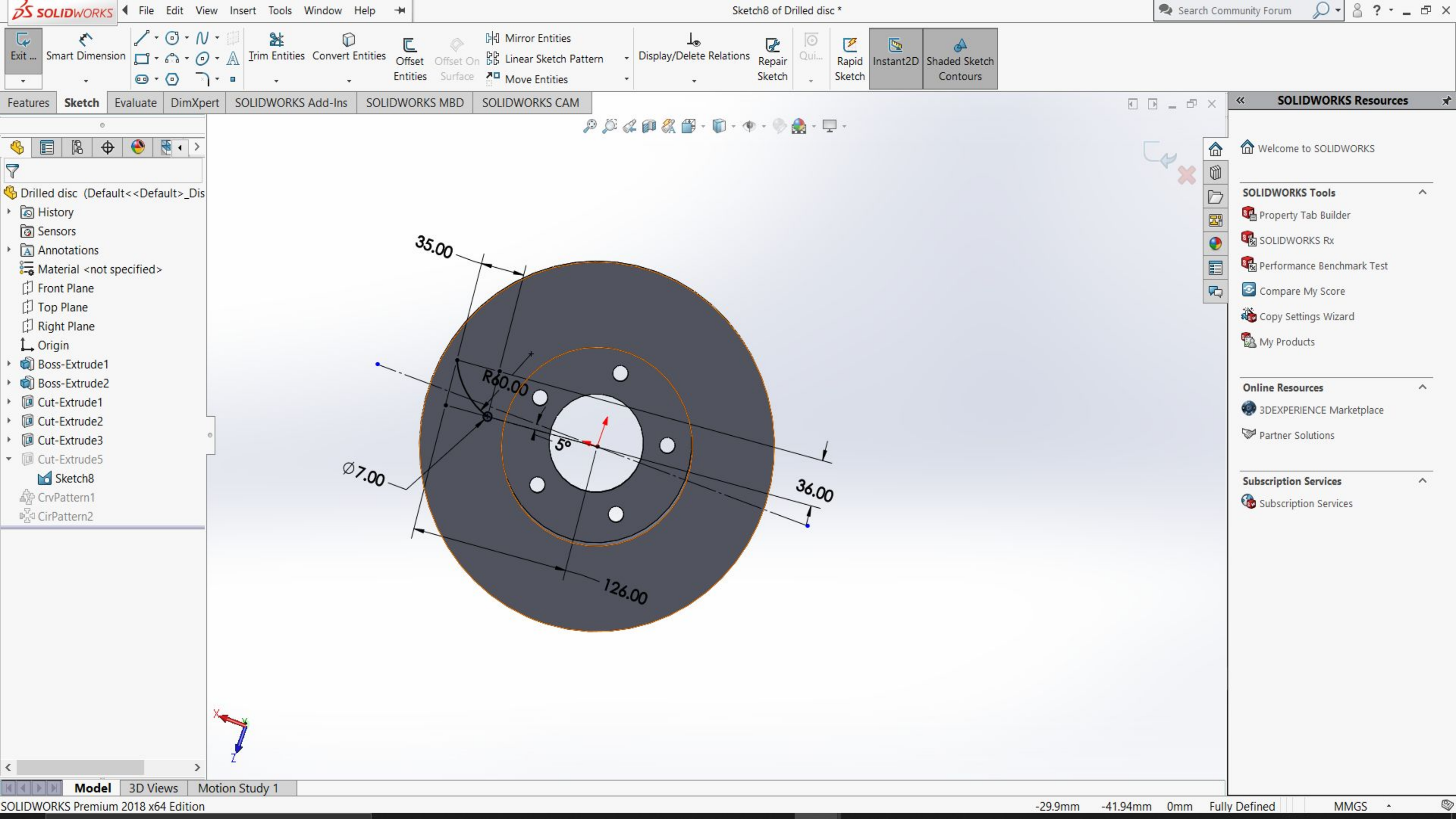
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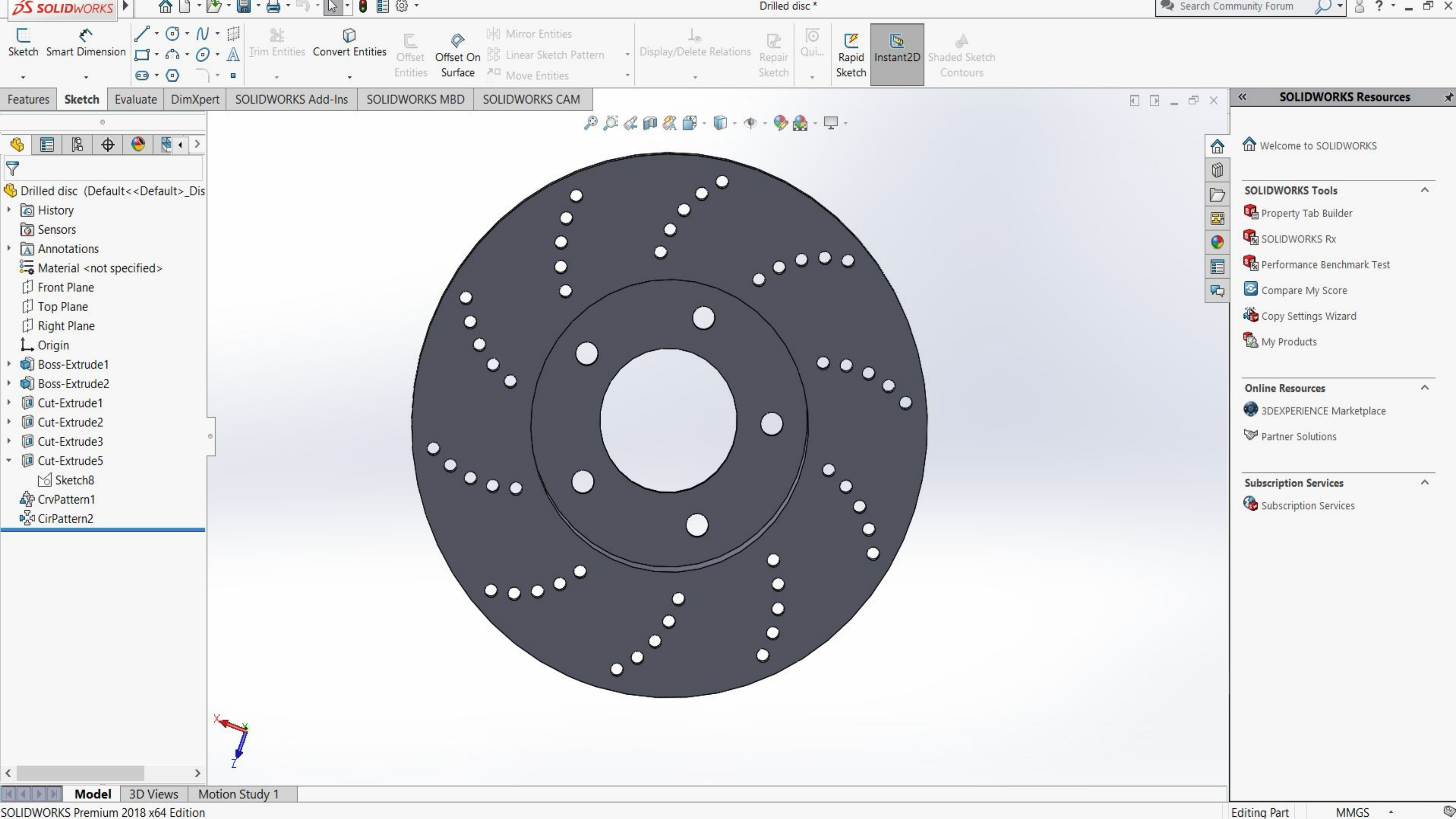
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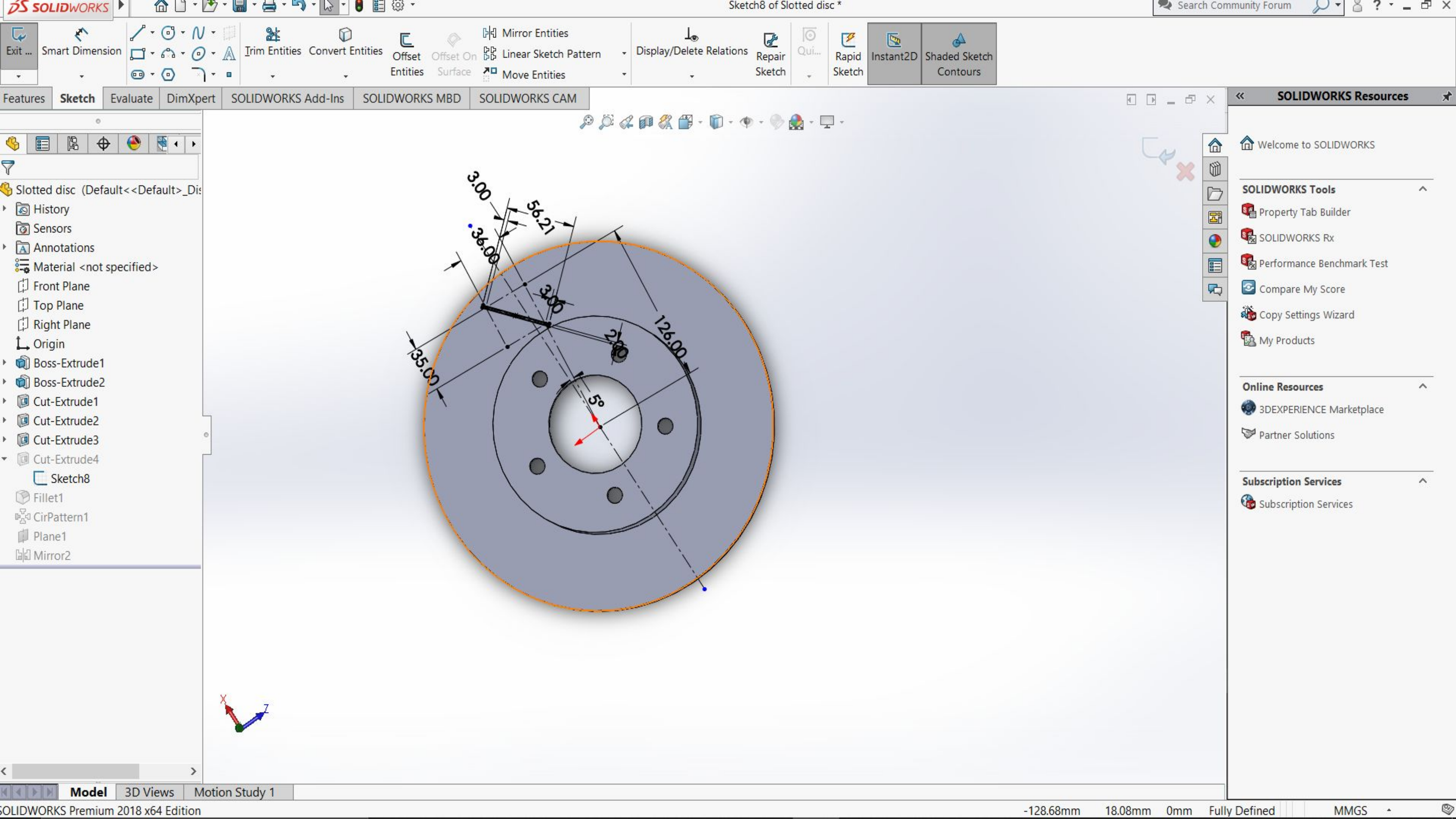
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Motion Study 1
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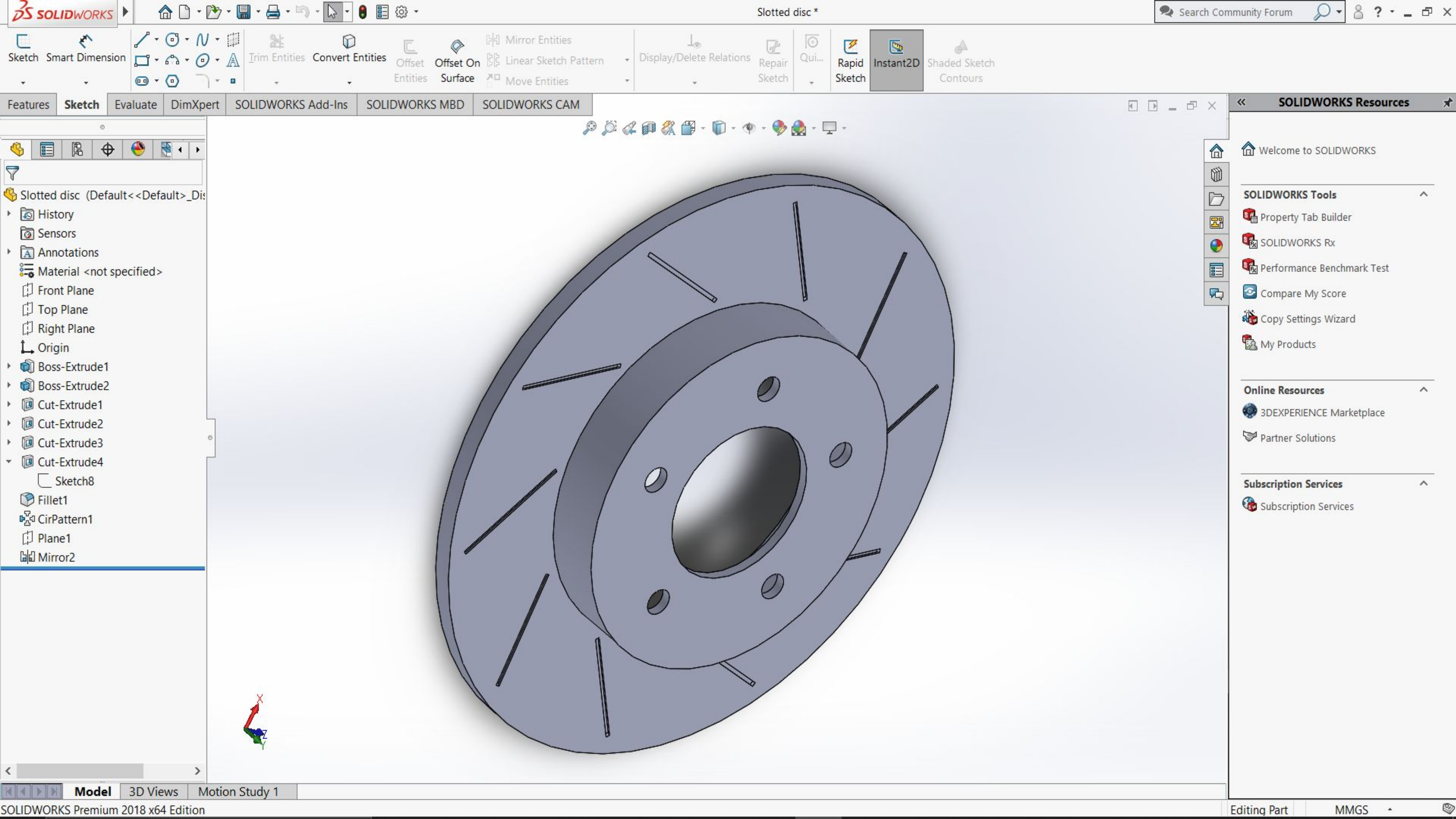
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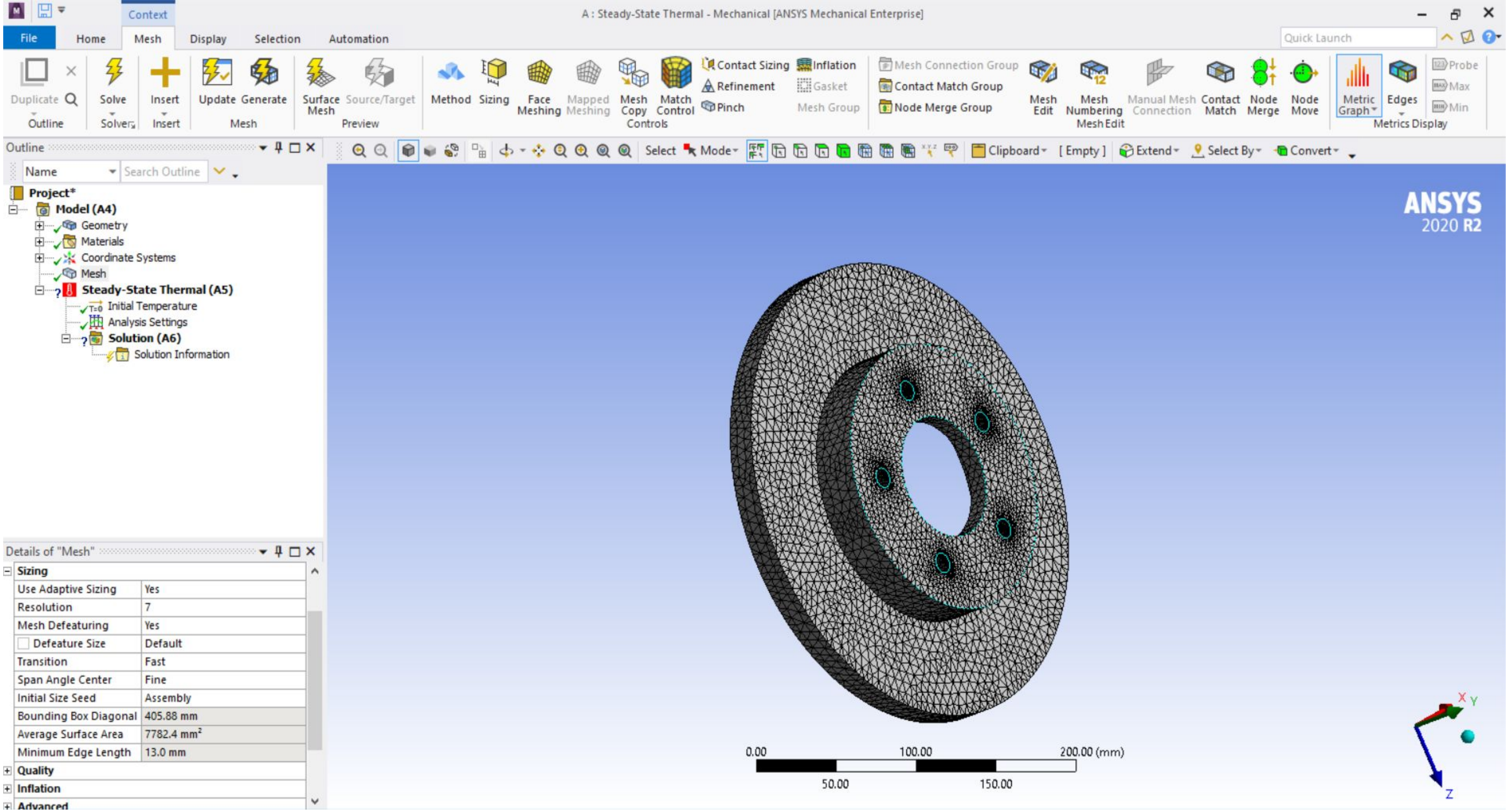
STEADY STATE THERMAL ANALYSIS

- Temperature variation and heat flux throughout the geometry of the rotor are calculated and analyzed here.
- Analysis was based on thermal loading and does not determine the life of the disc rotor.
- Ambient air cooling is taken into account and forced convection is applied.
- Boundary conditions were calculated by referring different research papers.
- Analysis of different type of disc rotors were done on Ansys Workbench.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Analysis Settings
Solution (A6)
Solution Information
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Average Surface Area 7782.4 mm²
Minimum Edge Length 13.0 mm
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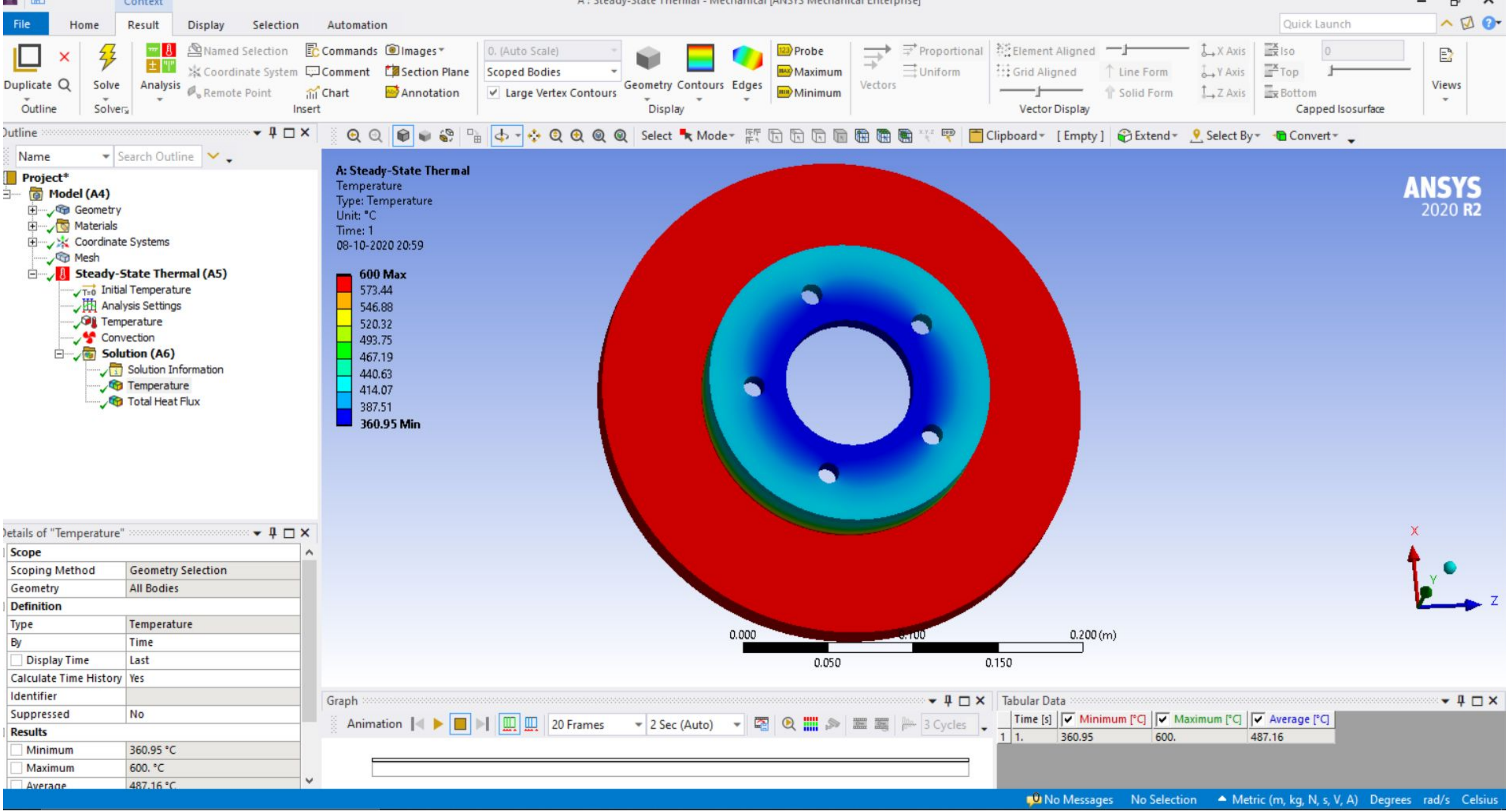
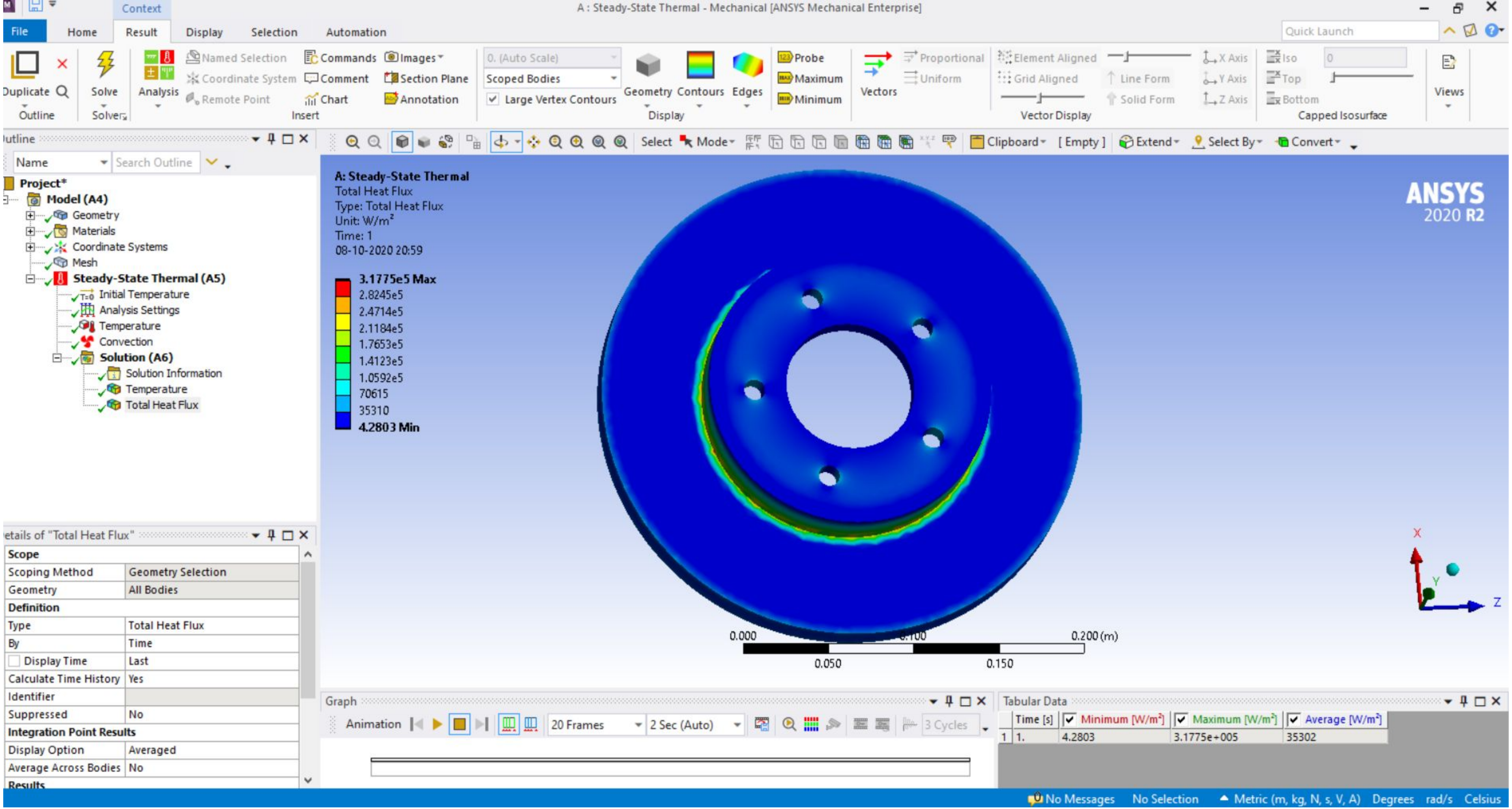
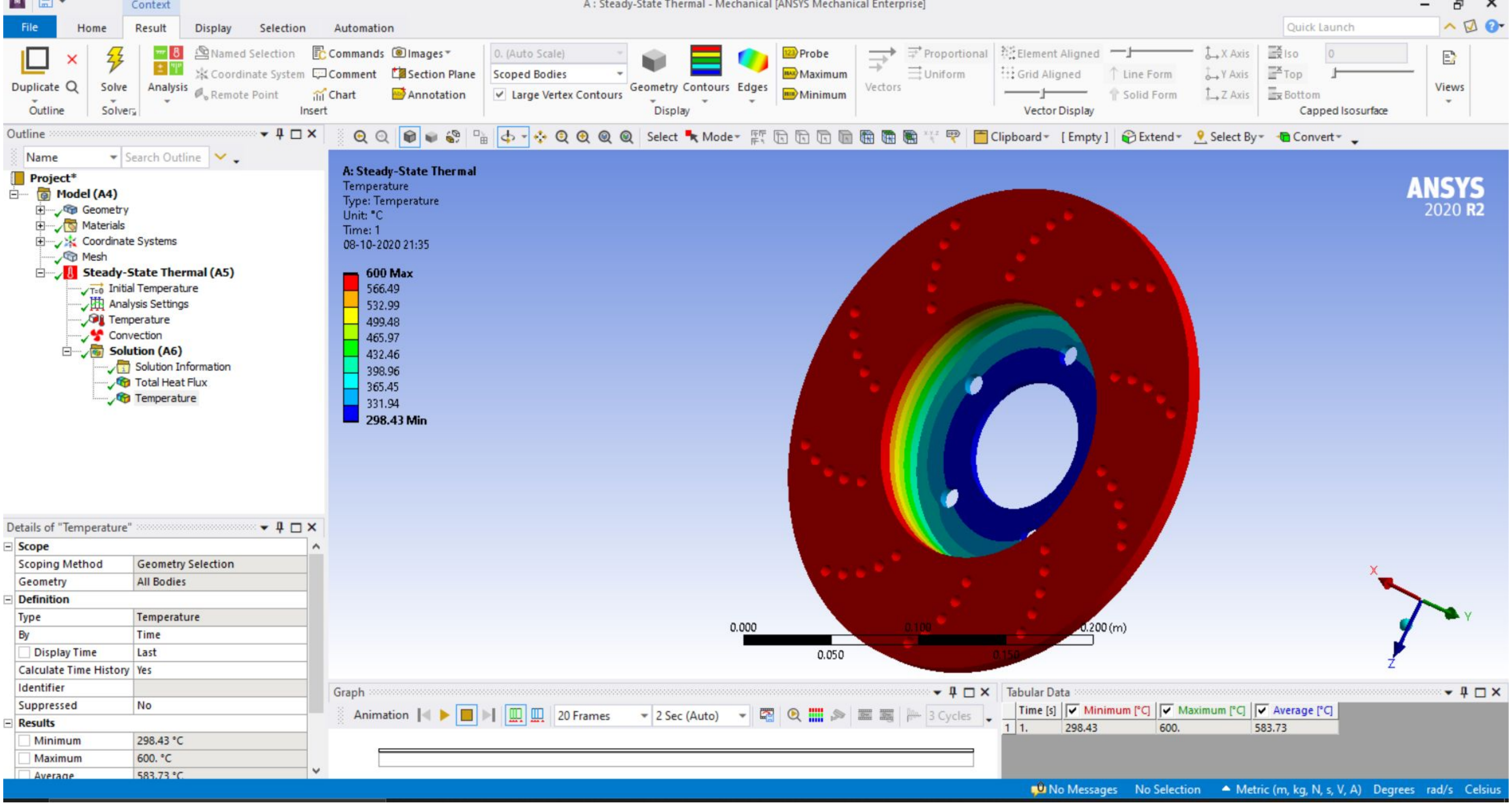


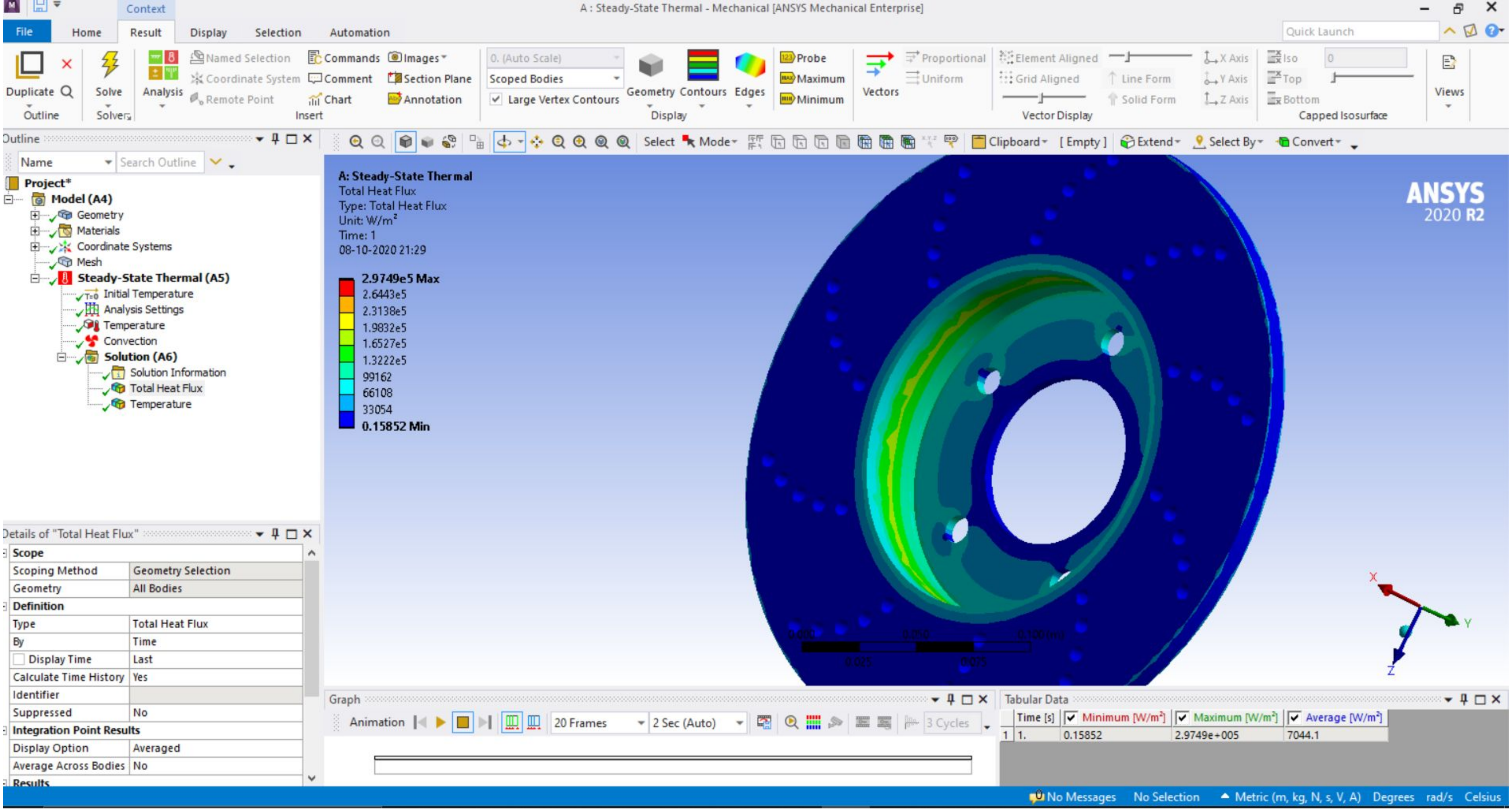
Temperature distribution over a solid disc rotor



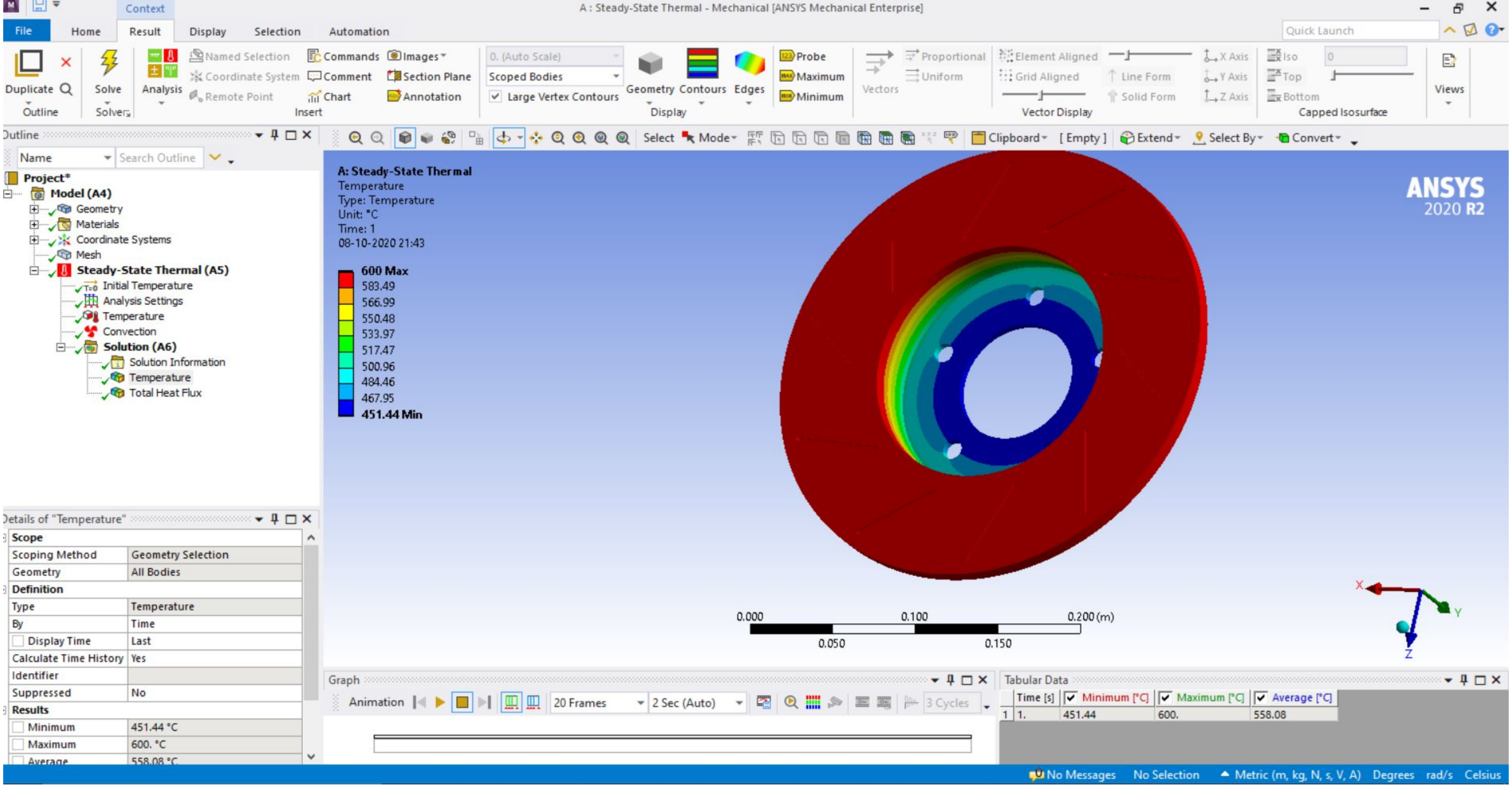
Total heat flux in a solid disc rotor



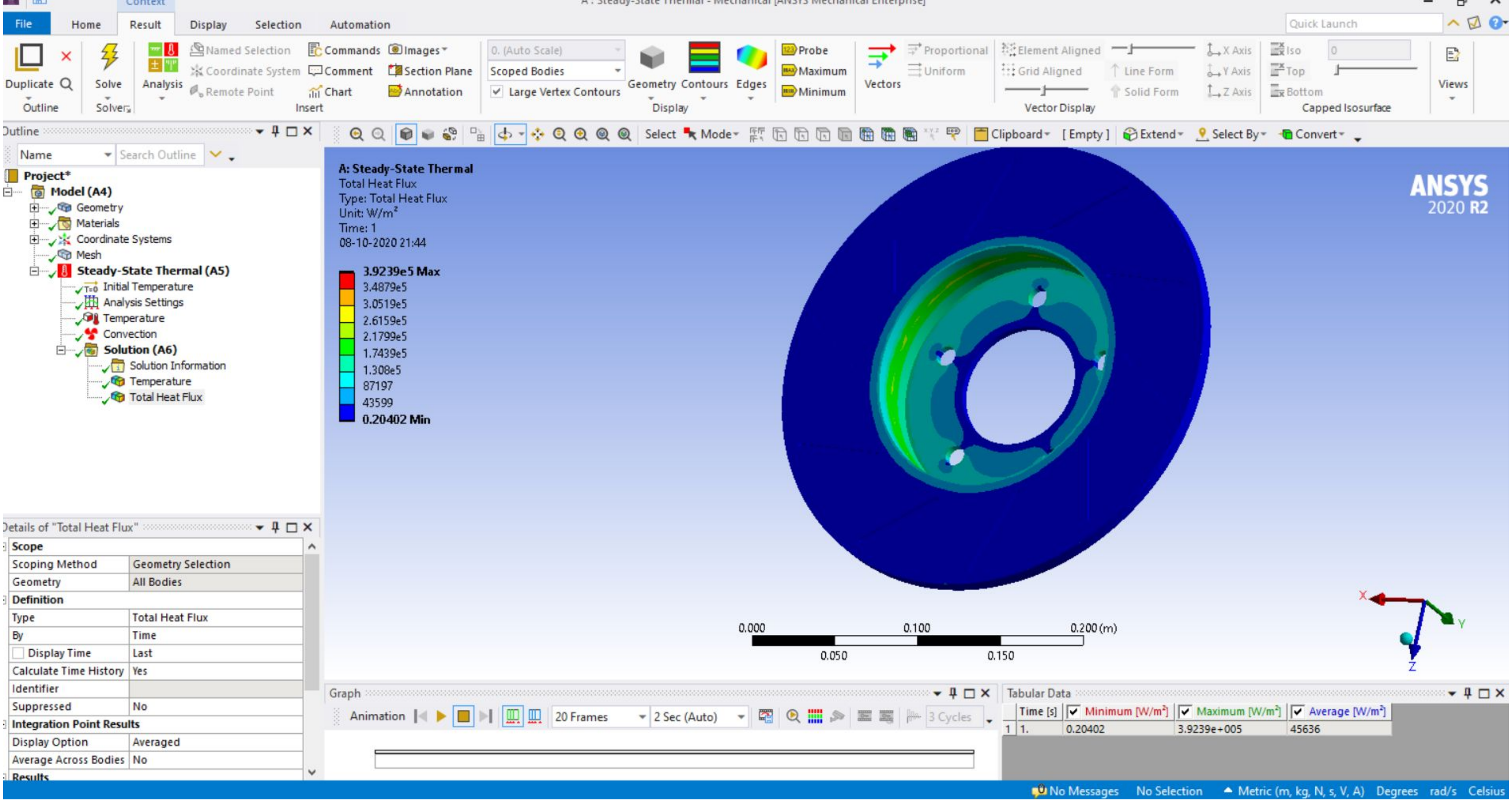
Temperature distribution over a drilled disc rotor



Total heat flux in a drilled rotor



Temperature distribution over a slotted disc rotor



Total heat flux in a slotted disc rotor

CONCLUSION

Solid, drilled and slotted disc of 285mm outer diameter and 167mm inner diameter with flange thickness of 39mm were made using Solidworks and analyzed on Ansys.

On the basis of the analysis and results, the heat flux and temperature distribution for the three rotors are nearly same. Even though the drilled disc rotor have a better value for both heat flux and temperature distribution due to increased surface area for heat dissipation due to drilled hole. For effective breaking the heat dissipation should be higher for that a higher surface area is required, drilling holes provides higher surface area but compromises the strength of the material. For have an efficient rotor there should have an optimization between thermal and structural characteristics of the disc rotor.

THANK YOU